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The AI Infrastructure Engineer Roadmap: Five Phases, 37 Weeks

A structured path from AI fundamentals to advanced distributed systems — compute, storage, networking, training, serving, observability, and HPC. With an Apple Silicon track woven through for engineers building on-device AI.

9 MIN READ · AI INFRASTRUCTURE · CAREER ROADMAP · LEARNING PATH

COMPUTE · STORAGE · NET | PYTORCH · DEEPSPEED | TRITON · RAY | HPC · RDMA | APPLE SILICON

Why This Roadmap, Why Now

Behind every production AI model is a stack most resumes barely touch: specialized accelerators, parallel file systems, low-latency interconnects, distributed training schedulers, inference servers, drift detectors, and compilers that turn graphs into kernels. **AI infrastructure engineering** is the discipline that owns that stack — and the demand for it has decisively outpaced supply.

What follows is a sequenced 37-week path through five phases. It is opinionated about order: foundations before infrastructure, infrastructure before training, training before serving, serving before observability, and observability before the deep cuts (distributed AI, HPC, custom GPU kernels). Each phase lists the courses worth doing, what you should walk away with, and how to surface the skill on a resume so it actually reads as infrastructure work rather than a generic "AI/ML" bullet.

An *Apple Silicon* add-on track runs in parallel from Phase 2 onward, for engineers building or shipping on-device AI — Core ML, Metal, Instruments, and the compiler chain that connects them.

At a Glance

PHASE	FOCUS	WEEKS	CUMULATIVE
1	Foundation & Mindset	4	4
2	AI Infrastructure Fundamentals	8	12
3	Model Training & Serving	10	22
4	AI Data & Observability	6	28
5	Advanced AI Systems	9	**37**

Phase 1 — Foundation & Mindset

Before any infrastructure, get the vocabulary and the ethics right. You need to be able to discuss model lifecycles, data flows, and failure modes with practitioners without translating every other word. Skip this phase and the rest reads like a list of disconnected tools.

Phase 1: Building the AI Bedrock

4 WEEKS

Terminology, ethics, the data → training → deployment lifecycle.

Elements of AI UNIVERSITY OF HELSINKI

GOAL Build AI fundamentals, terminology, ethics, and a feel for real-world applications.

RESUME Add an "AI fundamentals" line showing grounding in core concepts and ethics.

Introduction to Artificial Intelligence COURSERA / LINKEDIN LEARNING

GOAL Refresh AI basics and the full lifecycle — data, training, deployment.

RESUME Strengthen "AI lifecycle knowledge"; highlight understanding of model deployment stages.

Phase 2 — AI Infrastructure Fundamentals

This is where you stop treating "the cluster" as a black box. AI workloads have specialized needs at every layer of the stack: GPUs and accelerators with their own memory hierarchies, parallel storage that can saturate them, networking that doesn't collapse under collectives, and IaC that can spin all of it up reproducibly. Compilers belong here too — once you see how MLIR and LLVM lower a model graph, a lot of inference performance work stops being magic.

Phase 2: The Backbone

8 WEEKS

Compute, storage, networking, IaC — the substrate AI workloads actually run on.

AI Infrastructure and Operations Fundamentals COURSERA

GOAL Understand AI-specific compute, storage, and networking needs.

RESUME Add AI compute stack and deployment architecture expertise under "Technical Skills."

AI-Driven Infrastructure as Code (IaC) and Cloud Automation UDEMY

GOAL Deploy AI infrastructure programmatically across cloud and on-prem.

RESUME Showcase IaC skills for AI infrastructure deployment under "Cloud & Automation."

Introduction to MLIR and LLVM for AI Model Compilation APPLE ADD-ON LLVM.ORG / MLIR.DEV

GOAL Understand compiler infrastructure for AI model optimization and transformation.

RESUME Add compiler and runtime optimization skills for on-device AI execution.

Phase 3 — Model Training & Serving

Hands-on time. By the end of this phase you should have trained or fine-tuned a model end-to-end, scaled a training job across multiple GPUs or nodes, and served a model behind a real inference server — ideally one that supports both batch and real-time workloads. The Apple add-on rounds this out with the on-device serving paths (Core ML, TFLite) that are increasingly the deployment target for production mobile AI.

Phase 3: Bringing AI to Life

10 WEEKS

Training, fine-tuning, distributed scaling, and inference serving.

Practical Deep Learning for Coders FAST.AI

GOAL Hands-on AI model training, fine-tuning, and GPU workflows.

RESUME Add hands-on AI model training and GPU workflows under "AI Engineering Projects."

Distributed Training with PyTorch and DeepSpeed UDEMY

GOAL Scale training jobs with multi-GPU, multi-node setups.

RESUME Highlight distributed model training expertise under "AI Infrastructure."

Deploying AI Models with NVIDIA Triton Inference Server UDEMY

GOAL Serve models efficiently in real-time and batch inference.

RESUME Add model serving and optimization with NVIDIA Triton under "AI Deployment."

Deploying Models to Core ML and TensorFlow Lite APPLE ADD-ON APPLE DEVELOPER / TENSORFLOW.ORG

GOAL Convert and deploy models for iOS/macOS devices using Core ML and TFLite.

RESUME Add mobile and on-device AI deployment with Core ML and TFLite.

Phase 4 – AI Data & Observability

Production AI breaks in ways traditional software doesn't. Data drifts. Model quality degrades silently. Latency spikes correlate with batch composition rather than load. This phase wires up the pipelines and dashboards that let you see those failures before users do – data versioning with DVC, metrics through Prometheus/Grafana, and on Apple devices, the Instruments-based profiling that surfaces energy and memory cost.

Phase 4: Reliability and Performance

6 WEEKS

Versioned data, traced pipelines, drift detection, and energy profiling.

Data Version Control (DVC) and ML Pipelines UDEMY

GOAL Version datasets, track experiments, and manage AI workflows.

RESUME Include data versioning and ML pipeline management under "Data Engineering."

Real-Time AI Monitoring with Prometheus and Grafana UDEMY

GOAL Monitor model drift, latency, and performance.

RESUME Add AI observability and drift detection metrics under “Monitoring.”

Performance Profiling and Energy Optimization on Apple Silicon APPLE ADD-ON

INSTRUMENTS / XCODE

GOAL Profile latency, memory, and energy usage for AI workloads on Apple devices.

RESUME Highlight on-device AI performance tuning with Instruments and Apple Silicon profiling.

Phase 5 – Advanced AI Systems

The last phase is where you stop being a generalist on the AI stack and start being credible on the hardest parts of it: distributed training and serving frameworks like Ray, the HPC layer (RDMA, InfiniBand, NCCL tuning) that determines whether your collectives scale or stall, and – on the Apple side – writing custom Metal kernels for performance work the framework can’t do for you.

Phase 5: Pushing the Boundaries

9 WEEKS

Ray, HPC networking, custom GPU kernels – the deep cuts.

Ray for Distributed AI UDEMY / OFFICIAL DOCS

GOAL Distributed AI training, hyperparameter tuning, and scaling.

RESUME Highlight distributed AI scaling and Ray under “Distributed Systems.”

High-Performance Computing for AI LINKEDIN LEARNING

GOAL Learn HPC concepts: RDMA, InfiniBand, NCCL tuning.

RESUME Show advanced HPC tuning and GPU networking under “Performance Optimization.”

Metal Performance Shaders and GPU Programming for AI APPLE ADD-ON APPLE DEVELOPER

GOAL Leverage GPU acceleration for AI models on Apple devices using Metal APIs.

RESUME Add Apple GPU optimization and custom AI kernel programming experience.

How to Actually Use This

A few things this roadmap will not do for you, and a few things it will:

- **Courses do not equal experience.** Each phase needs at least one small project that lives in a public repo — a Triton deployment, a DVC pipeline, a Ray-on-Kubernetes lab. The course gets you to the starting line; the project is what you actually talk about in interviews.
- **The order matters.** Phase 5's NCCL tuning makes no sense without Phase 3's distributed training and Phase 2's networking model. Resist the urge to jump ahead because a topic sounds more impressive.
- **The Apple track is optional but coherent.** If you are not shipping on-device AI, you can skip the Apple add-ons without losing the spine of the roadmap. If you *are* shipping on-device AI, the four add-ons (MLIR, Core ML/TFLite, Instruments, Metal) form their own mini-curriculum.
- **37 weeks is a calendar estimate, not a contract.** Engineers with adjacent backgrounds — systems, distributed databases, HPC — will move faster through Phases 2 and 5. Engineers coming from application development will likely need longer in Phases 3 and 4.

The shape of the role is clear: infrastructure people who understand the AI workloads running on top, and AI people who understand the systems running underneath. This roadmap is one path to the intersection. The work itself — the projects, the on-call, the cluster reading at 2am — is what turns it into a career.

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